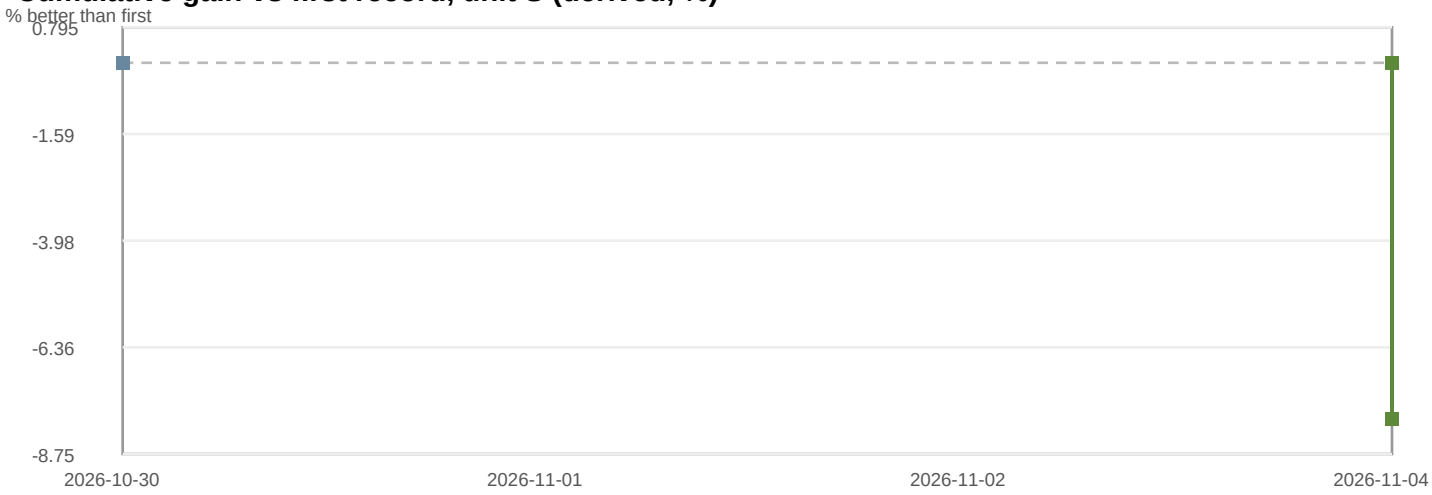
Benchmark progress: cryptanalysis

From bench.record/v1 records in Git; raw data in S3 by SHA-256. Derived numbers are labelled derived.

Records: open 0, measured 11, invalid 0, censored 0. Headline unit S: group operations / sqrt(n); the ECDLP cost unit, every phase priced.

Cumulative gain vs first record, unit S (derived, %)



Series key

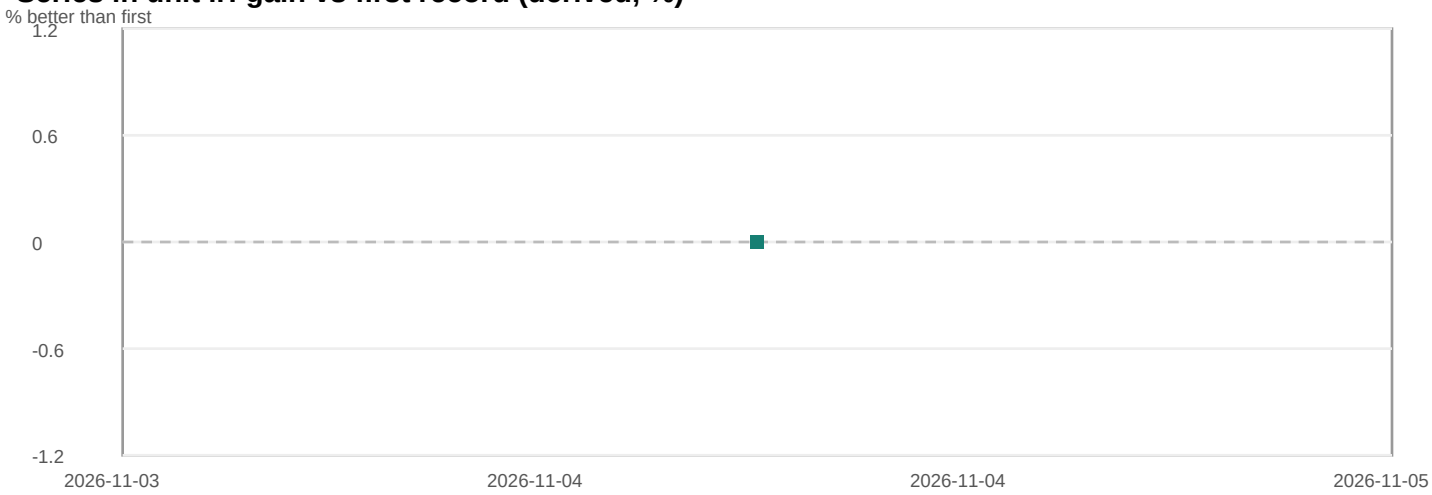
- ecdlp:EC1N83Ce1h35c7ced6b497 •
- ecdlp:EC1N83Ce1h35c7ced6b497 •
- ecdlp:ca-bench-generic-ec • bs
- ecdlp:ca-bench-generic-ec • gr
- ecdlp:ca-bench-generic-ec • ka
- ecdlp:ca-bench-generic-ec • rh
- ecdlp:ca-bench-generic-ec • rh
- ecdlp:rho-plain-negmap • mean_
- ecdlp:rho-plain-negmap • mean_
- ecdlp:rho-plain-negmap • mean_
- ecdlp:rho-plain-negmap • mean_
- ecdlp:rho-plain-negmap • mean_
- ecdlp:rho-plain-negmap • mean_

Distance remaining: value / reference (1.0 = parity; derived)

Series key

- ecdlp:EC1N83Ce1h35c7ced6b497 •
- ecdlp:EC1N83Ce1h35c7ced6b497 •
- ecdlp:ca-bench-generic-ec • bs
- ecdlp:ca-bench-generic-ec • gr
- ecdlp:ca-bench-generic-ec • ka
- ecdlp:ca-bench-generic-ec • rh
- ecdlp:rho-plain-negmap • mean_
- ecdlp:rho-plain-negmap • mean_
- ecdlp:rho-plain-negmap • mean_
- ecdlp:rho-plain-negmap • mean_
- ecdlp:rho-plain-negmap • mean_
- ecdlp:rho-plain-negmap • mean_

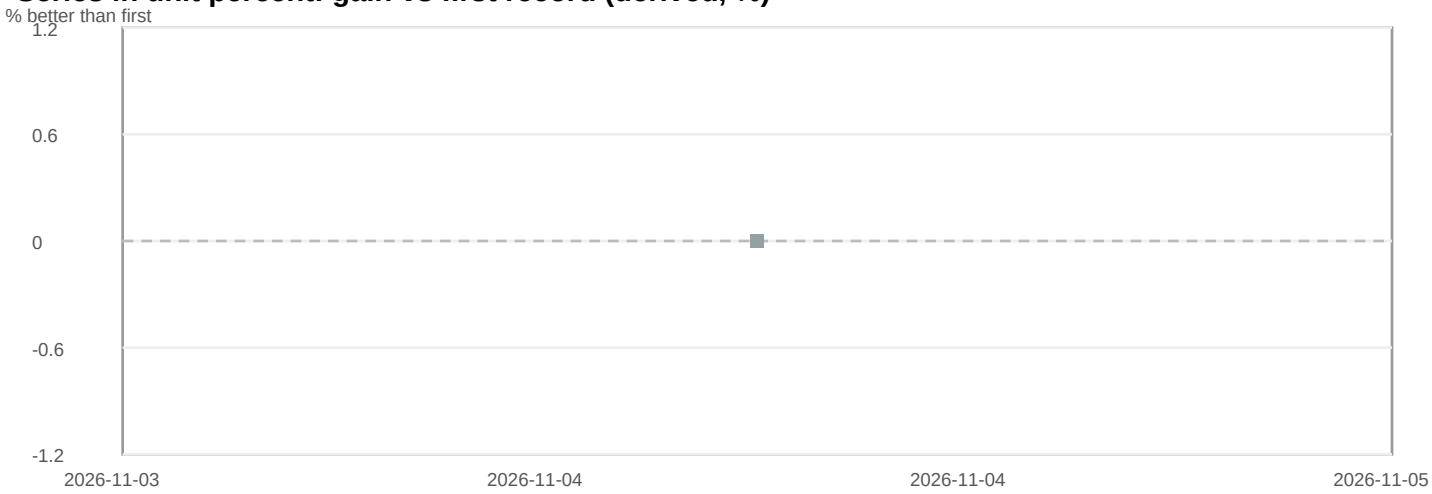
Series in unit ir: gain vs first record (derived, %)



Series key

- ic:N53 • ir.koblitz_descent_k0
- ic:N53 • ir.koblitz_rung_k0n31
- kernel:suite_gf2_koblitz • ir.
- kernel:suite_gf2_koblitz • ir.
- kernel:suite_gf2_koblitz • ir.
- kernel:suite_gf2_koblitz • ir.
- kernel:suite_gf2_koblitz • ir.
- kernel:suite_gf2_koblitz • ir.
- kernel:suite_gf2_koblitz • ir.
- kernel:suite_gf2_koblitz • ir.

Series in unit percent: gain vs first record (derived, %)



Series key

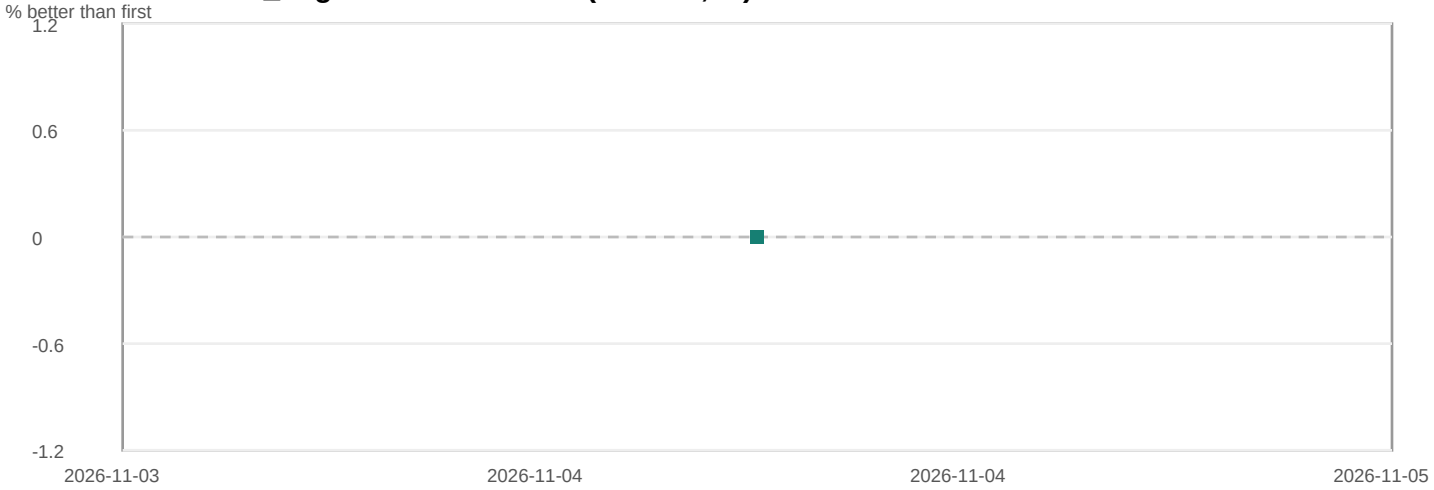
- ic:N53 • filter_admit_rate
- ic:N53 • share.contains_key_lo
- ic:N53 • share.add_many_lazy
- ic:N53 • share.keys_of

Series in unit ratio: gain vs first record (derived, %)

Series key

- ecdlp:EC1N83Ce1h35c7ced6b497 •
- ecdlp:EC1N83Ce1h35c7ced6b497 •
- ecdlp:EC1N83Ce1h35c7ced6b497 •
- ecdlp:EC1N83Ce1h35c7ced6b497 •
- ecdlp:EC1N83Ce1h35c7ced6b497 •
- ecdlp:rho-plain-negmap • group
- ecdlp:rho-plain-negmap • group
- ecdlp:rho-plain-negmap • group
- ecdlp:rho-plain-negmap • group
- ecdlp:rho-plain-negmap • group
- ecdlp:rho-plain-negmap • group
- ecdlp:rho-plain-negmap • group
- ic:N53 • wall_ratio.koblitz_de
- ic:N53 • wall_ratio.koblitz_ru
- kernel:gf2_n53 • wall_ratio.gf
- kernel:gf2_n53 • wall_ratio.ko
- kernel:suite_gf2_koblitz • per
- kernel:suite_gf2_koblitz • wal
- kernel:suite_gf2_koblitz • wal
- kernel:suite_gf2_koblitz • wal
- kernel:suite_gf2_koblitz • wal
- kernel:suite_gf2_koblitz • wal
- kernel:suite_gf2_koblitz • wal
- kernel:suite_gf2_koblitz • wal

Series in unit wall_s: gain vs first record (derived, %)



Series key

- ecdlp:EC1N83Ce1h35c7ced6b497 •

Series summary

problem • metric	unit	first	latest	gain vs first	value/ref	pts
ecdlp:EC1N83Ce1h35c7ced6b497 • e2e_cold_wall_b	wall_s	8.976	8.976	0%	4.0949	1
ecdlp:EC1N83Ce1h35c7ced6b497 • e2e_paired_a_over	ratio	1.261	1.261	-0%	-	1
ecdlp:EC1N83Ce1h35c7ced6b497 • s_e2e_cold	S	0.3922	0.3922	0%	4.0955	1
ecdlp:EC1N83Ce1h35c7ced6b497 • s_online	S	0.0075	0.0075	0%	-	1
ecdlp:EC1N83Ce1h35c7ced6b497 • stage_index_build	ratio	1.3069	1.3069	-0%	-	1
ecdlp:EC1N83Ce1h35c7ced6b497 • stage_online_a_ov	ratio	1.4053	1.4053	-0%	-	1
ecdlp:EC1N83Ce1h35c7ced6b497 • stage_rank_stage_	ratio	1.1996	1.1996	-0%	-	1
ecdlp:EC1N83Ce1h35c7ced6b497 • stage_scan_selfte	ratio	1.3165	1.3165	-0%	-	1
ecdlp:ca-bench-generic-ec • bsgs_s_ec40	S	1.5729	1.5729	0%	0.9183	2
ecdlp:ca-bench-generic-ec • grumpy_s_ec40	S	1.2514	1.2514	0%	0.7306	2
ecdlp:ca-bench-generic-ec • kangaroo_s_ec40	S	1.2201	1.2201	0%	0.7123	2
ecdlp:ca-bench-generic-ec • rho1_s_ec40	S	1.7129	1.7129	0%	-	2
ecdlp:ca-bench-generic-ec • rhot_s_ec40	S	1.2415	1.3402	-7.9508%	0.7824	2
ecdlp:rho-plain-negmap • group_ops_ratio_20	ratio	0.6661	0.6661	0%	-	1
ecdlp:rho-plain-negmap • group_ops_ratio_24	ratio	0.6296	0.6296	0%	-	1
ecdlp:rho-plain-negmap • group_ops_ratio_28	ratio	0.5875	0.5875	0%	-	1
ecdlp:rho-plain-negmap • group_ops_ratio_32	ratio	0.8612	0.8612	0%	-	1
ecdlp:rho-plain-negmap • group_ops_ratio_36	ratio	0.8856	0.8856	0%	-	1
ecdlp:rho-plain-negmap • group_ops_ratio_40	ratio	0.9345	0.9345	0%	-	1
ecdlp:rho-plain-negmap • group_ops_ratio_pooled	ratio	0.7483	0.7483	0%	-	1
ecdlp:rho-plain-negmap • mean_s_20	S	1.441	1.441	0%	0.6653	1
ecdlp:rho-plain-negmap • mean_s_24	S	1.149	1.149	0%	0.6289	1
ecdlp:rho-plain-negmap • mean_s_28	S	1.033	1.033	0%	0.5876	1
ecdlp:rho-plain-negmap • mean_s_32	S	0.98	0.98	0%	0.8604	1
ecdlp:rho-plain-negmap • mean_s_36	S	0.95	0.95	0%	0.8854	1
ecdlp:rho-plain-negmap • mean_s_40	S	0.909	0.909	0%	0.9342	1
ic:N53 • filter_admit_rate	percent	12.29	12.29	0%	-	1
ic:N53 • ir.koblitz_descent_k0n53_m2_x4	ir	5.73e+08	5.73e+08	0%	-	1
ic:N53 • ir.koblitz_rung_k0n31	ir	2.026e+08	2.026e+08	0%	-	1
ic:N53 • share.add_many_lazy	percent	38.5	38.5	0%	-	1
ic:N53 • share.contains_key_loop	percent	35.3	35.3	0%	-	1
ic:N53 • share.keys_of	percent	10.6	10.6	0%	-	1
ic:N53 • trials	count	1.207e+07	1.207e+07	-	-	1
ic:N53 • wall_ratio.koblitz_descent_k0n53_m2_x4	ratio	1.254	1.254	-0%	-	1
ic:N53 • wall_ratio.koblitz_rung_k0n31	ratio	1.151	1.151	-0%	-	1
kernel:gf2_n53 • wall_ratio.gf2_n53_mul_sqr_1m	ratio	0.794	0.794	-0%	-	1
kernel:gf2_n53 • wall_ratio.koblitz_collect_k0n5	ratio	0.865	0.865	-0%	-	1
kernel:suite_gf2_koblitz • ir.gf2_n53_inv_16k	ir	1.978e+07	1.978e+07	0%	-	1
kernel:suite_gf2_koblitz • ir.koblitz_collect_k0	ir	3.147e+08	3.147e+08	0%	-	1
kernel:suite_gf2_koblitz • ir.koblitz_collect_k0	ir	3.261e+08	3.261e+08	0%	-	1
kernel:suite_gf2_koblitz • ir.koblitz_curve_mul_	ir	5.401e+07	5.401e+07	0%	-	1
kernel:suite_gf2_koblitz • ir.koblitz_descent_k0	ir	7.312e+08	7.312e+08	0%	-	1
kernel:suite_gf2_koblitz • ir.koblitz_rung_k0n31	ir	2.314e+08	2.314e+08	0%	-	1
kernel:suite_gf2_koblitz • ir.koblitz_signed_rho	ir	3.364e+08	3.364e+08	0%	-	1
kernel:suite_gf2_koblitz • perfindex_wall_index	ratio	1.2148	1.2148	-0%	-	1
kernel:suite_gf2_koblitz • wall_ratio.gf2_n53_in	ratio	4.1436	4.1436	-0%	-	1
kernel:suite_gf2_koblitz • wall_ratio.koblitz_co	ratio	1.1933	1.1933	-0%	-	1
kernel:suite_gf2_koblitz • wall_ratio.koblitz_co	ratio	1.0116	1.0116	-0%	-	1
kernel:suite_gf2_koblitz • wall_ratio.koblitz_cu	ratio	1.4638	1.4638	-0%	-	1
kernel:suite_gf2_koblitz • wall_ratio.koblitz_de	ratio	1.2539	1.2539	-0%	-	1
kernel:suite_gf2_koblitz • wall_ratio.koblitz_ru	ratio	1.1078	1.1078	-0%	-	1
kernel:suite_gf2_koblitz • wall_ratio.koblitz_si	ratio	1.1986	1.1986	-0%	-	1

Every value is quoted from the record it names, in that record’s unit. Percentages and ratios under ’derived’

are computed here from those values and are not measurements. Series in different units are never compared. 3 digest-pinned historical measured records are excluded from all series and derived comparisons; recorded status counts retain them. See integrity_exclusions in progress.json and PROGRESS.md for reasons.

Historical records excluded from comparisons

benchmarks/records/20261104T0352Z-descent-walks-k0n53.json: Record includes failed commands without a machine-readable command-to-metric attribution. Conservative aggregate exclusion retains successful and failed attempts; review individual raw outputs before comparison.

benchmarks/records/20261104T1632Z-pair-filter-two-bit.json: Record includes failed commands without a machine-readable command-to-metric attribution. Conservative aggregate exclusion retains successful and failed attempts; review individual raw outputs before comparison.

benchmarks/records/20261105T0736Z-rho-plain-negmap-retrospective.json: Documentary retrospective commands have no captured successful exit status. Values remain historical quotations; execution provenance is incomplete.